Agentic Travel Planning: Where AI Helped, and Where It Did Not

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Abstract

Many agentic tasks begin with an open-ended goal but are useful only when hard constraints are met. Multi-day travel planning is a popular example for such a task. We describe our entry to the "2nd Travel Planning Challenge with Agentic AI" at the IJCAI 2026 conference[1]. Curiously, we found that the best-performing system was a deterministic, environment-grounded planner that completely ignores the available large language model (LLM). Instead, it uses classical optimization and acceptance gates that can only emit itineraries confirmed valid against the environment rules and APIs.

Conversely, AI agents were most valuable in building and improving that system: We successfully used the competition as a low-risk environment to run a near-continuous data-driven multi-agent AI development process, generating hundreds of commits over eight weeks. The agents proposed experiments, implemented changes, ran evaluations, and reviewed pull requests, while humans made every decision that mattered: what to prioritize and how to split the work, how to measure progress and provide a fast, meaningful feedback loop for the agents, and what to finally submit.

1 Introduction

Many real-world tasks involve a loosely stated human goal that needs to be turned into an output that is not merely plausible but correct. For example, multi-day travel planning combines fuzzy human requirements that must be turned into exact constraints for the constructed itinerary to fulfill. The IJCAI 2026 Travel Planning Challenge[1] extends the ChinaTravel benchmark of Shao et al. [Shao *et al.*, 2024]: for a given human travel intent, an agent must produce a preference-aligned itinerary between a selection of Chinese cities. A request might say:

> "Plan three days in Suzhou, visit the Humble Administrator's Garden between 9 and 11, stay near the old town, avoid taxis, and spend at most CNY 3,000."

[1]Website: https://chinatravel-competition.github.io/IJCAI2026/, Benchmark: https://github.com/LAMDA-NeSy/ChinaTravel (URLs accessed on August 9, 2026)

The output is structured data, not free text. Hotels, attractions, and restaurants must exist in the provided sandbox environment; opening hours, transportation, and costs must be consistent; logical requirements must hold; and commonsense rules such as meal counts must pass. The evaluator measures environmental validity (EPR), logical satisfaction (C-LPR and FPR), and preferences. Hard constraint satisfaction heavily dominates the overall score – preference terms only count if all hard requirements are fulfilled (else set to zero).

The competition has two phases. In the offline first phase, released verifier fields and a generous time budget allowed us to satisfy almost all constraints and then polish preferences. In the harder phase 2, which we focus on in this report, only query text and environment tools are exposed to the agent. For each query, the agent must infer what constraints to satisfy and how – within 5 hours total for 100 unknown queries.

Our system follows a propose–verify–repair pattern related to LLM-Modulo systems [Kambhampati *et al.*, 2024], but assigns roles asymmetrically: an optional LLM may interpret language or propose changes; environment APIs establish facts; and deterministic optimization and checks decide what may be emitted. Our final configuration disables the LLM path.

2 System Design

2.1 Pipeline and trust boundary

Figure 1 follows the example from language to a simplified itinerary. Oracle verifier fields are removed before execution (only available in the public data, not the held-out test set). The only inputs are visible query text and allowed environment tools. Every point of interest (POI), hotel, restaurant, price, opening hour, and transportation record is read from `WorldEnv`; nothing is generated as a plausible substitute.

2.2 Parsing and normalization

The parser extracts dates, cities, budgets, excluded transportation, named POIs, timed visits, hotel properties, room counts, cuisines, and attraction categories. It also handles alternatives ("visit A *or* B") by creating separate solver variants rather than an impossible conjunction. Grounding and normalization resolve names and concepts against each city's catalog. Deterministic extraction is fast and testable but

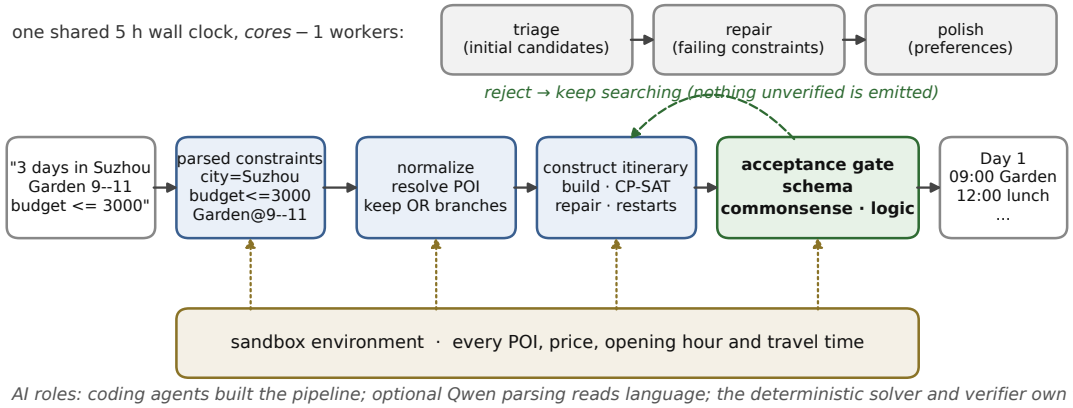


Figure 1: The per-query pipeline and whole-run scheduler, illustrated with the opening example. Optional model-based extraction may fill missing constraints; grounding, construction, repair, and acceptance remain deterministic.

not synonymous with understanding, so we added phrasing-robust fallbacks for the largest measured recall gaps and an optional LLM[2] extractor that may fill a missing family but never overwrite a deterministic value. Section 4 compares this path with the deterministic extractor and explains why it is disabled in our final agent.

Precision is equally important. Our audit found that in 23–50% of free-prose queries, depending on the set, the parser initially invented a required POI name that did not exist in the target city. Such a query has no feasible solution at all, so it can never earn a full-plan point no matter how long the solver runs. Therefore, a catalog check in the final parser removes such provably unsatisfiable required names and category values, after first trying to salvage a real POI name from the captured text. It reduced this measured failure class to zero on every set we hold.

Further investigation illustrates how narrow these parsing gaps are. Until a recent fix to our final agent, intercity transport was the largest remaining unextracted family on the free-prose dataset: the extractor recognized "take the train to the destination" but not "we have chosen to fly" or "by high-speed train (G)", missing 33 of 85 such requirements on one released set and 27 of 32 on another. Adding verb-form and qualified-noun patterns, guarded so that a station name or an undecided "fly or take the train" never produces a requirement, removed most of them while leaving precision and the familiar register unchanged.

2.3 Construction, repair, and densification

The deterministic planner constructs a complete base itinerary from grounded candidate pools. It then applies bounded CP-SAT [Perron and Didier, 2024], domain-specific repairs, and seeded restarts. Repairs include placing named activities inside requested windows, inserting absent required POIs, closing transportation gaps, and rebuilding intercity legs. The same acceptance checker evaluates every candidate: query constraints, schema, and commonsense rules must all pass. Accepted candidates are ranked by preference value and cost.

Packing and densification add meals and activities only while the checker keeps passing. A minimal schema-valid itinerary is written before solving so an interrupted run still has output — availability, not success, since an empty itinerary scores zero.

2.4 Scheduling the five-hour run

The whole-split scheduler uses *triage*, *repair*, and *polish* passes in parallel across queries: triage reaches every query cheaply, repair revisits the nearest failures with larger budgets, and polish spends what is left only after hard requirements pass. A keep-best rule makes updates monotonic, and deadline checks prevent starting work that cannot finish. As summarized in Figure 1, triage quickly constructs an accepted candidate for every query; repair re-solves near misses with increasing per-query budgets; and polish optimizes preferences for passing plans. Each accepted improvement is written immediately, so a limited budget or interrupt leaves the best plan found.

3 Evaluation

We evaluate three questions: what official feedback says about generalization, what changed fro our v1 to v2 submission, and whether deterministic planning outperforms model-based baselines. We call the released template-style queries *familiar* and the differently worded extensions to the public ChinaTravel dataset *free prose*[3].

3.1 Official feedback

In the final, second phase of the competition, the organizers permit two submitted harness versions and retain the better score. We submitted v1 early to obtain the only available held-out feedback before choosing v2. On the held-out set, our overall score was 33.25 versus the leader's 36.41.

[2]The organizers provide a Qwen3.6-27B model for evaluation.

[3]https://huggingface.co/datasets/LAMDA-NeSy/ChinaTravel, explicitly referenced by the organizers.

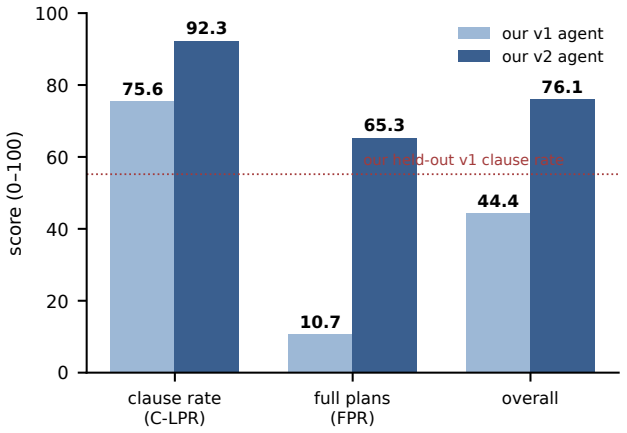


Figure 2: Submitted v1 versus the second version on 150 unfamiliar-phrasing queries, paired at equal budget; higher is better. The dotted line is our v1 clause rate on the organizers' held-out set, a different query set, shown for scale. At the time of writing, our v2 agent is prepared but not submitted, so its held-out score is unknown.

Grounding generalized: micro EPR—the share of itinerary fields consistent with the environment—remained 99.48, within 0.16 of the best entry. Constraint understanding did not: C-LPR—the share of requested clauses satisfied—fell from 90.08 on familiar queries to 55.20 held out. No query passed every requirement, so all preference terms were zero.

Across the five submitted systems, four had a full-plan rate of zero and the fifth had one point; every preference term was zero. Thus full-plan failure was not unique to us, but our C-LPR drop pointed specifically to parsing and motivated the v2 extraction work below. The entire 3.16-point gap to the leader corresponds to 12.6 clause points.

3.2 What the second version changes

In the final two days before the deadline, we repaired the phrasing brittleness described below, removed the unsatisfiable-name failure class, and closed the largest remaining recall gaps. None of that is visible on the familiar register, where the extractor was already complete, so we measured it where it applies: 150 differently phrased released queries, both builds run as a paired experiment at identical budget on one host, scored against an English-literal oracle so the full-plan rate has a real denominator.[4] Figure 2 shows the result: the clause rate rises from 75.6 to 92.3 and the full-plan rate from 10.7 to 65.3. Checked clause by clause, 225 of 1264 clauses move from failing to passing against 13 lost, and the number of queries satisfying every requirement rises from 16 to 98 of 150. On the familiar register, where most of this cannot help, a paired run of the full five-hour pipeline confirms no regression and a modest gain: 64.69 to 66.81 overall, with the full-plan rate moving from 45 to 49 of 100 queries.

[4]Those released sets state their query text in English but their verifier literals in Chinese, which caps the achievable full-plan rate; we translate the literals with an audited one-to-one map for measurement only, and the agent never reads them.

The largest single fix illustrates a measurement lesson. Our coverage audits classified the hotel room count as "guaranteed by trip metadata", because the planner derives it from party size and beds per room — correct only when the traveler says nothing. Asking a different question, *for each query that fails exactly one requirement, which requirement is it?*, showed that 28 of 150 free-prose queries failed *only* the room count: one field away from satisfying everything. Reading the stated count instead of deriving it moved the full-plan rate on that set from 48% to 65%. No coverage table could have found this; a near-miss analysis found it in minutes.

3.3 Query phrasing

The familiar released set was presented as in distribution, so prioritizing it was a reasonable choice. Its queries use a regular numbered template, whereas other released sets express the same requirements in varied prose: offline audits found no missed clauses on the familiar register but 12–29% on the differently phrased ones. We had optimized against the one register we were already perfect on.

3.4 Planning baselines

We ran two small, matched comparisons. An autonomous tool-using Qwen agent scored 1.88 and 2.64 on eight familiar and eight free-prose queries, versus 67.71 and 48.49 for our solver; the upstream RuleNeSy baseline [Shao *et al.*, 2024] scored about 9.5 on four of each. Neither produced an itinerary passing every commonsense rule. These samples are too small to rank systems, but the failure modes support retaining deterministic acceptance and optimization: on these samples our solver scored 18–36 times higher than autonomous Qwen and about five to seven times higher than RuleNeSy.

4 Where AI Helped — and Where It Did Not

4.1 A safe vibe-coding experiment

We used this challenge deliberately as a low-risk environment in which to push agent-driven development ("vibe coding") much harder than we would on a production system. With an objective scorer, nothing deployed, and no external dependencies, we could ask how far coding agents can carry a real system given fast feedback.

Very far, as it turned out, given a great deal of scaffolding and proper initial design. We ran multiple paid agent subscriptions across several machines, with one or more agents working almost continuously for eight weeks. The result is hundreds of commits, 100+ reviewed and merged pull requests, more than 1000 tests, and dozens of long-running evaluation measurements. Humans still made every decision that mattered: what to prioritize and how to split the work, how to measure progress and provide a fast, meaningful feedback loop for the agents, and what to finally submit. Three practices worked well:

1. Split the roles by cost and by kind of thinking. Expensive reasoning models were used for analysis, diagnosis, and planning: reading measurements, proposing the next experiment, ranking work by expected effect. Faster, cheaper agents then implemented a specified change with tests, and

ran evaluations. The planning agent maintained a ranked work list; implementation and evaluation proceeded in parallel branches against it, preventing implementing agents from silently redefining their task. Cheaper agents could ask a stronger advisor for guidance.

2. Make feedback fast, automatic, and cheap to trust. We invested more in the evaluation harness than in any single planner feature: a paired A/B screen that decides in one or two hours whether a change is worth a long run, a whole-run mode reproducing the submission's exact shape, deterministic seeds, and result ledgers recording every condition needed to decide whether two numbers may be compared. Agents could then evaluate their own work overnight and report a verdict against a criterion written before the run. We split available data into validation and held-out sets to limit overfitting.

3. Write the invariants down and let them be enforced by tests. Grounding, no oracle leakage, acceptance-before-emission, and evaluation proof before enabling new features are project rules pinned by tests, so an agent that would violate one fails rather than persuades. Cold-start verification caught a start-up race that would have shipped empty itineraries for a whole split while still exiting successfully.

Two limits are worth stating. The volume of agent output is not itself a result: much of it was superseded, and the ratio of committed work to accepted improvement was correspondingly low. Agents were also often over-optimistic about their changes, making rigorous evaluation and promotion essential.

4.2 Inside the system

We tested the LLM in three distinct roles: parsing, repair, and end-to-end planning. For these experiments, a DeepInfra API provided the same Qwen3.6-27B model version used by the organizers without a local deployment. LLM-first extraction cut missing free-prose constraint families by 40–65%, but initially converted many misses into values that did not match the environment vocabulary. We therefore connected model output to the same city-specific entity and category catalogs used by the solver. Grounded, fill-only extraction improved one small familiar A/B test but was flat on the matched free-prose subset, and still matched fewer clauses than the contemporaneous deterministic parser. Our final agent consequently makes zero LLM calls, while retaining the grounded extractor as an optional path. This is not a rejection of LLM parsing in general; it separates recognizing a requirement from resolving its values against the actual database.

We retain the optional parser because implicit categories that no pattern names still suit a language model, provided its answer is resolved against the closed vocabulary.

We also tried a hybrid: letting a model repair itineraries the checker rejected, with every proposed edit re-verified before it could be kept. On ten deterministic failures the loop iterated, proposed valid-looking edits, and flipped none of them to passing. That's why we did not enable it in the final system.

End-to-end itinerary generation was the clearer boundary. Even with environment tools, autonomous Qwen planning accumulated small inconsistencies, and one violation invalidated the whole itinerary. Dates, opening windows, travel times, budgets, meals, and disjunctions remain a constrained optimization problem, where deterministic solving provides reproducibility, explicit acceptance, and efficient targeted repair.

5 Conclusion

The challenge rewarded a hybrid division of labor. Coding agents accelerated construction and analysis; the itinerary agent itself is deterministic and makes zero LLM calls. Environment tools supply facts, deterministic search assembles them, and acceptance gates admit only verified plans. Official v1 feedback showed that grounding transferred while extraction tuned to familiar phrasing did not; our v2 submission measurably repairs that gap.

The transferable lesson is methodological: agentic coding works well in an isolated project with fast, objective feedback. Written invariants, comparable measurements, and independent promotion runs mattered more than output volume. Models may help at the grounded language–symbol boundary, but our measured model path did not beat deterministic extraction, and classical optimization remained better for itinerary construction.

If we wanted to extend our travel planning agent to become a production system, we would replace the sandbox environment with live map, hotel, rail, flight, and booking APIs, and would infer user-specific objectives rather than inherit fixed challenge preferences: some travelers want four attractions per day, others value unplanned time. Live availability would require replanning, but the same division holds: AI interprets intent, APIs establish reality, optimization constructs the itinerary, and deterministic verification decides what to emit.

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